
Examples Reference:<https://adjectivesproject.org/?page=data>

Chapter 1 Affine Schemes

1.1 Locally Ringed Spaces

Note The knowledge here should be listed in the part of sheaves. But the locally ringed is mainly used in algebraic geometry. It is convenient to introduce here.

Definition 1.1.0.1

- A **locally ringed space** is sheaf $(X, \mathcal{O}_X)$, whose stalks $\mathcal{O}_{X,x}$, $x \in X$ are local rings.
- Define the **residue field**

$$\kappa(x) := \mathcal{O}_{X,x}/m_x$$

- The morphism of locally ringed spaces is the morphism of ringed spaces

$$(f, f^\#) : (X, \mathcal{O}_X) \rightarrow (Y, \mathcal{O}_Y)$$

via this lemma: ?? There exists a correspondence (It is actually an f -map)

$$f^\# : f^{-1}\mathcal{O}_Y \rightarrow \mathcal{O}_X \quad f^b : \mathcal{O}_Y \rightarrow f_*\mathcal{O}_X$$

- Composition: Consider

$$f : X \rightarrow Y \quad g : Y \rightarrow Z$$

Then, there exists a composition

$$\mathcal{O}_{Z,g(f(x))} \xrightarrow{g^\#} \mathcal{O}_{Y,f(x)} \xrightarrow{f^\#} \mathcal{O}_{X,x}$$

Then, $(f, f^\#)$ induces the morphism of stalks

$$f_x^\# : \mathcal{O}_{Y,f(x)} \rightarrow \mathcal{O}_{X,x}$$

which is a local ring map. It maps $m_{Y,f(x)}$ to $m_{X,x}$.

It is easy to check, stalks isomorphism $\Leftrightarrow$ Local ring isomorphisms.

Definition 1.1.0.2 Trivial: Functor

Locally ringed spaces $\rightarrow$ Ringed Spaces

which reflects isomorphisms.

1.1.1 Morphisms

Open Immersion of Locally Ringed Spaces

Definition 1.1.1.1 Let $f : X \rightarrow Y$ be a morphism of locally ringed spaces. We say that f is an **open immersion** if f is homeomorphism of X onto an open subset of Y , and the map

$$f^\# : f^{-1}\mathcal{O}_Y \rightarrow \mathcal{O}_X$$

is an isomorphism.

♣ **Example 1.1.1.2** Consider the open subset $U \subseteq X$, then

$$j : (U, \mathcal{O}_U) \rightarrow (X, \mathcal{O}_X)$$

is an open immersion. Where $\mathcal{O}_U(V) := \mathcal{O}_X(V)$; $V \subseteq U$.

Definition 1.1.1.3 Let X be a locally ringed space. $U \subseteq X$ is a closed subset. Then, the scheme above is called an **open subspace** of scheme $(X, \mathcal{O}_X)$.

◇ **Lemma 1.1.1.4** Let $f : X \rightarrow Y$ be an open immersion of locally ringed spaces with $j : V = f(X) \rightarrow Y$ be the open subspace of Y associated to the image of f . Then: There exists unique isomorphism $f' : X \cong V$ of locally ringed spaces such that

$$f = j \circ f'$$

◦ **Proof** Firstly, let $f' : X \rightarrow V = f(X)$ the homomorphism between X and V induced by f . Then we have $f = j \circ f'$, where f' is the homeomorphism.

Then, there exists an isomorphism of sheaves

$$f^\# : f^{-1}\mathcal{O}_Y \rightarrow \mathcal{O}_X$$

because of open immersion, there exists an isomorphism of rings

$$f^\# : \Gamma(U, f^{-1}\mathcal{O}_Y) \rightarrow \Gamma(U, \mathcal{O}_X)$$

And we have $f^{-1}\mathcal{O}_Y \cong f'^{-1}\mathcal{O}_V$:

$$\mathcal{O}_V = j^{-1}\mathcal{O}_U \Rightarrow f^{-1}\mathcal{O}_Y = (j \circ f')^{-1}\mathcal{O}_Y = f'^{-1}(j^{-1}\mathcal{O}_Y) = f'^{-1}\mathcal{O}_V$$

thus we have an isomorphism $\Gamma(U, (f')^{-1}\mathcal{O}_V) \rightarrow \Gamma(U, \mathcal{O}_X)$. Now we've seen an isomorphism

$$f'^{-1}(\mathcal{O}_V) \rightarrow \mathcal{O}_X$$

Consider $(f', f'^\# : f'^{-1}\mathcal{O}_V \rightarrow \mathcal{O}_X)$.

Uniqueness: Suppose there exists another $f'' : (X, \mathcal{O}_X) \rightarrow (V, \mathcal{O}_V)$, where

$$f' = f'' \in Top$$

then

$$\begin{array}{ccc} \Gamma(U, f^{-1}\mathcal{O}_Y) & \xrightarrow{\cong} & \Gamma(U, \mathcal{O}_X) \\ \downarrow \cong & \searrow (f')^\# & \uparrow (f'')^\# \\ \Gamma(U, (f')^{-1}\mathcal{O}_V) & & \end{array}$$

The locally ringed space is unique. □

◇ **Lemma 1.1.1.5** Let $f : X \rightarrow Y$ be a morphism of locally ringed spaces. Let $U \subseteq X$, and $V \subseteq Y$ be open subsets. Suppose $f(U) \subseteq V$. There exists a unique morphism of locally ringed spaces $f|_U : U \rightarrow V$ such that the following diagram is a commutative square of locally ringed spaces

$$\begin{array}{ccc} U & \longrightarrow & X \\ \downarrow f|_U & & \downarrow f \\ V & \longrightarrow & Y \end{array}$$

◦ **Proof** Omitted. □

There exists a factorization

$$X \rightarrow V \rightarrow Y$$

where $f|_U$ is an open immersion.

Closed Immersion of Locally Ringed Spaces

Definition 1.1.1.6 Let $i : Z \rightarrow X$ be a morphism of locally ringed spaces. We say that i is a **closed immersion** if:

- i is a homeomorphism of Z onto a closed subset of X ;
- $\mathcal{O}_X \rightarrow i_*\mathcal{O}_Z$ is surjective; Denote

$$\mathcal{I} := \text{Ker}(\mathcal{O}_X \rightarrow i_*\mathcal{O}_Z)$$

- $\mathcal{I}$ is locally generated by sections.

Remark Sometimes, the third condition

$\mathcal{I}$ is locally generated by sections

is not necessary. (Especially, for the theory of geometry).

The main problem is

- Partition of Unity;
- Bump functions;

In complex, we have the following theorem

► **Theorem** (Oka's Coherence Theorem) The sheaf of holomorphic functions $\mathcal{O}_X^{an}$ on complex is a coherent sheaf.

♣ **Example 1.1.1.7** Objects in $C^\infty(\mathbb{R})$: Consider

$$\mathcal{I}(U) := \{f \in C^\infty(U) | f(x) = 0, \forall x \in U \cap [0, \infty)\}$$

- Homeomorphism: $[0, \infty)$;
- Surjectivity: Whitney's extension theorem
https://etd.ohiolink.edu/acprod/odb_etd/ws/send_file/send?accession=ucin1712912147070133&disposition=inline
- Locally generated: Not hold. Consider the neighborhood of 0.

◇ **Lemma 1.1.1.8** Let $f : Z \rightarrow X$ be a morphism of locally ringed spaces.

In order for f to be a closed immersion, it suffices that there exists an open covering $X = \cup U_i$ such that

$$f^{-1}U_i \rightarrow U_i$$

is a closed immersion.

◦ **Proof** ... □

Sheaf of Ideals

Now, we consider the sheaf ideals, as $\mathcal{O}_X$ submodule, which is locally generated by sections. Now, the support

$$Z := \text{supp}(\mathcal{O}_X/I) = \{x \in X \mid (\mathcal{O}_X/I)_x \neq 0\}$$

is closed, with inclusion $i : Z \rightarrow X$. There exists a unique sheaf of rings $\mathcal{O}_Z$ on Z , with

$$i_*\mathcal{O}_Z := \mathcal{O}_X/I$$

Then, its stalk

$$(i_*\mathcal{O}_Z)_x = (\mathcal{O}_X/I)_x = \mathcal{O}_{X,i(z)}/I_{i(z)}$$

Closed Subspace

Definition 1.1.1.9 Let X be a locally ringed space, I be a sheaf of ideals on X , which is locally generated by sections.

Then, the **locally ringed space** is the **closed subspace** of X , associated to the sheaf of ideals I .

◇ **Lemma 1.1.1.10** Let $f : X \rightarrow Y$ be a closed immersion of locally ringed spaces. Let

$$I := \text{Ker}(\mathcal{O}_Y \rightarrow f_*\mathcal{O}_X)$$

Let $i : Z \rightarrow Y$ be the closed subspace associated to I . There exists a unique isomorphism $f' : X \cong Z$ of locally ringed spaces such that

$$f = i \circ f'$$

◦ **Proof ...** □

◇ **Lemma 1.1.1.11** Let X, Y be locally ringed spaces. $I \subseteq \mathcal{O}_X$ be a sheaf of ideals, locally generated by sections.

Let $i : Z \rightarrow X$ be the associated closed subspace.

Then, a morphism $f : Y \rightarrow X$ factors through $Z \Leftrightarrow$ the map $f^*I \rightarrow f^*\mathcal{O}_X = \mathcal{O}_Y$ is zero.

If this is the case the morphism $g : Y \rightarrow Z$ such that $f = i \circ g$ is unique.

◦ **Proof ...** □

♠ **Proposition 1.1.1.12** Let $f : X \rightarrow Y$ be a morphism of locally ringed spaces, $I \subseteq \mathcal{O}_Y$ be a sheaf of ideals, which is locally generated by sections.

Let $i : Z \rightarrow Y$ be the closed subspace associated to the sheaf of ideals I , and

$$\mathcal{J} := \text{Im}(f^*I \rightarrow f^*\mathcal{O}_Y = \mathcal{O}_X)$$

Then, $\mathcal{J}$ is ideal locally generated by sections.

Moreover, let $i' : Z' \rightarrow X$ be the associated closed subspace of X , there exists a unique morphism of locally ringed spaces

$$f' : Z' \rightarrow Z$$

such that the following diagram is a commutative locally ringed spaces

$$\begin{array}{ccc} Z' & \xrightarrow{i'} & X \\ \downarrow f' & & \downarrow f \\ Z & \xrightarrow{i} & Y \end{array}$$

Moreover, this diagram is a **fibre square** in the category of locally ringed spaces.

◦ **Proof**

- Locally generated:
- Factorization:

□

♠ **Proposition 1.1.1.13** Given a subsheaf $\mathcal{I}$ of $\mathcal{O}_X$. Let $Z = \text{supp}(U \mapsto \mathcal{O}_X(U)/\mathcal{I}(U))$. Then, we have the isomorphism

$$\mathcal{O}_X/\mathcal{I} \cong i_*\mathcal{O}_Z$$

if $\mathcal{I}$ is locally generated.

◦ **Proof** Now, for any $x \in X$:

- $x \notin Z \Rightarrow \mathcal{O}_X(U)/\mathcal{I}(U) = 0 \Rightarrow \mathcal{O}_{X,x} = \mathcal{I}_{X,x}$;
- $x \in Z \Rightarrow \mathcal{O}_{Z,x} := \mathcal{O}_{X,x}/\mathcal{I}_x$

The isomorphism

$$i_*\mathcal{O}_{Z,x} = \mathcal{O}_{X,x}/\mathcal{I}_x$$

By checking definition.

□

The quotient of stalk is also a local ring. So $(Z, \mathcal{O}_Z)$ is a locally ringed space.

1.2 Affine Schemes

1.2.1 Modules Sheaf

◇ **Lemma 1.2.1.1** Let R be a ring, $f \in R$:

- If $g \in R$ and $D(g) \subseteq D(f)$ then:
 - f is invertible in R_g ;
 - $g^e = af$ for some $e \geq 1, a \in R$;
 - Remark** Equivalently, $f \in \sqrt{(g)}$;
 - There is a canonical ring map $R_f \rightarrow R_g$;
 - There exists a canonical R_f -module map $M_f \rightarrow M_g$ for any R -module M ;
- (Every Open is Compact) Any open covering $D(f)$ can be refined to a finite open covering of the form

$$D(f) = \cup_{i=1}^n D(g_i)$$

- If $g_1, \dots, g_n \in R$, then $D(f) \subseteq \cup D(g_i) \Leftrightarrow g_i$ generate the unit ideal in R_f .

◦ **Proof**

- With $D(g) \cong \text{Spec}(R_g)$. Thus f is invertible in R_g . And $f \cdot \frac{a}{g^d} = 1 \Leftrightarrow (fg^d - a) \in \text{Ann}(g^{e-d}) \Rightarrow g^e = af$.

Map $R_f \rightarrow R_g, \frac{b}{f^n} \mapsto \frac{a^n b}{g^{ne}}$, or induced by inclusion $\text{Spec}(R_g) \hookrightarrow \text{Spec}(R_f)$.

Take the tensor product.

- Omitted. In CA.
- Omitted. In CA.

□

Definition 1.2.1.2 Let R be a ring:

- A **standard open covering** of $\text{Spec}(R)$ is a covering

$$\text{Spec}(R) = \bigcup_{i=1}^n D(f_i); f_i \in R$$

Remark Topology of $\text{Spec}(R)$ is compact, take the standard open covering.

- Suppose $D(f) \subseteq \text{Spec}(R)$ is a standard open. A **standard open covering** of $D(f)$ is a covering

$$D(f) = \bigcup_{i=1}^n D(g_i); g_i \in R$$

1.2.2 Modules Sheaf

Definition 1.2.2.1 Let R be a ring, and M be an R -sheaf. We will define a presheaf $\hat{M}$ on the basis of standard opens.

Suppose that $U \subseteq \text{Spec}(R)$ is a standard open. If $f, g \in R$ are such that $D(f) = D(g)$, then there are canonical maps

$$M_f \rightarrow M_g \quad M_g \rightarrow M_f$$

Then, define: For $U = D(f)$,

$$\hat{M}(U) = M_f$$

Suppose $D(g) \subseteq D(f)$, then there exists a canonical map

$$\hat{M}(D(f)) = M_f \rightarrow M_g = \hat{M}(D(g))$$

which is a presheaf of abelian groups.

Stalks

Take some $p = x \in \text{Spec}(R)$, we have

$$\hat{M}_x = \text{colim}_{f \in R, f \notin p} M_f$$

With the preorder property: Set $\{f \in R, f \notin p\}$ is preordered by the rule

$$f \geq f' \Leftrightarrow D(f) \subseteq D(f')$$

Via definition, we have: For $f_1, f_2 \in R - p$, we have

$$f_1 f_2 \subseteq f_1$$

By algebra we could finally conclude, that

$$\hat{M}_x = \text{colim}_{f \in R, f \notin p} M_f = M_p$$

By the contents of sheaf, there exists a correspondence

$$\text{Mod}(\mathcal{O}_X) \Leftrightarrow \text{Mod}(\mathcal{O}|B)$$

where B is a basis. So for the standard opens, $\mathcal{O}_{\text{Spec}(R)}$ is unique.

Sheaf Property

If $D(f) = \cup_{i=1}^n D(g_i)$, then the sheaf condition for this covering is equivalent to the exactness:

$$0 \longrightarrow M_f \longrightarrow \oplus M_{g_i} \longrightarrow M_{g_i g_j}$$

Notice that: $D(g_i) = D(f g_i)$. Hence rewritten as

$$0 \longrightarrow M_f \longrightarrow \oplus M_{f g_i} \longrightarrow \oplus M_{f g_i g_j}$$

This sequence is exact:??.

Definition 1.2.2.2 Let R be a ring:

- The **structure sheaf** $\mathcal{O}_{\text{Spec}(R)}$ of the spectrum of R is the unique sheaf rings $\mathcal{O}_{\text{Spec}(R)}$ which agrees with $\hat{R}$ on the basis of standard opens;
- The locally ringed space $(\text{Spec}(R), \mathcal{O}_{\text{Spec}(R)})$ is called the **spectrum** of R denoted by $\text{Spec}(R)$;
- The sheaf of $\mathcal{O}_{\text{Spec}(R)}$ -modules extending $\hat{M}$ to all open subsets on $\text{Spec}(R)$ is called the **sheaf of** $\mathcal{O}_{\text{Spec}(R)}$ -modules associated to M . This sheaf is denoted $\hat{M}$ as well.

◇ **Lemma 1.2.2.3** Let R be a ring, M be an R -module. Let $\hat{M}$ be the sheaf of $\mathcal{O}_{\text{Spec}(R)}$ -modules associated to M :

- $\Gamma(\text{Spec}(R), \mathcal{O}_{\text{Spec}(R)}) = R; \Gamma(\text{Spec}(R), \tilde{M}) = M$;
- $\Gamma(D(f), \mathcal{O}_{\text{Spec}(R)}) = R_f, f \in R; \Gamma(D(f), \tilde{M}) = M_f, f \in R$;
- Restriction maps for $D(g) \subseteq D(f)$

$$R_f \rightarrow R_g \quad M_f \rightarrow M_g$$

- Stalk: $\mathcal{O}_{\text{Spec}(R), x} = R_p, \tilde{M}_x = M_p$
- In particular,

$$R\text{-Mod} \rightarrow \text{Mod}(\mathcal{O}_{\text{Spec}(R)}) \quad M \mapsto \tilde{M}$$

is an exact functor.

◦ Proof

- Consider $D(1)$;
- Definition;
- Omitted;
- Omitted;
- We could prove this stalkwise: Notice that, the localization functor is exact, so the sequence in $\text{Mod}(\mathcal{O}_X)$ is exact on each $D(f)$. Thus the functor is exact.

□

1.2.3 The Structure Sheaf

Definition 1.2.3.1 Firstly, let A be a ring with $X = \text{Spec}(A)$. Define the **structure sheaf**

$$\mathcal{O}_X(U) := \{s : U \rightarrow \sqcup_{p \in U} A_p \mid s \text{ satisfies (i) (ii)}\}$$

with:

- For all $p \in U$, we have $s(p) \in A_p$;
- For all $p \in U$, there exists $a, b \in A$ and $V \subseteq U$, with $p \in V \subseteq D(b)$ and

$$s(q) = \frac{a}{b} \in A_q \quad q \in V$$

and $\rho_{UV}(s) = s|_V$.

Remark $b \in A - q$.

Remark There exists a natural ring homomorphism

$$\begin{aligned} A &\rightarrow \mathcal{O}_{\text{Spec}(A)}(\text{Spec}(A)) \\ a &\mapsto (p \mapsto \frac{a}{1} \in A_p) \end{aligned}$$

Definition 1.2.3.2 An **affine scheme** is a locally ringed space $(X, \mathcal{O}_X)$ if there exists a ring A , and an isomorphism

$$(X, \mathcal{O}_X) \xrightarrow{\sim} (\text{Spec}(A), \mathcal{O}_{\text{Spec}(A)})$$

1.2.4 AffSch and CRing Correspondence

◇ **Lemma 1.2.4.1** Let X be a locally ringed space. And $Y = \text{Spec}(A)$ be an affine scheme. $f : X \rightarrow Y$ be a morphism of locally ringed spaces.

Given a point $x \in X$ and consider the following ring maps

$$\Gamma(\mathcal{O}_Y, Y) = A \xrightarrow{f^\#} \Gamma(\mathcal{O}_X, X) \longrightarrow \mathcal{O}_{X,x}$$

Then, the preimage of m_x is prime in A , denoted by p , correspondent to $y \in \text{Spec}(A)$. We have

$$f(x) = y$$

◦ **Proof**

$$\begin{array}{ccc} \Gamma(X, \mathcal{O}_X) & \longrightarrow & \mathcal{O}_{X,x} \\ \uparrow & & \uparrow \\ \Gamma(Y, \mathcal{O}_Y) & \longrightarrow & \mathcal{O}_{Y,y} = A_p \end{array}$$

Take the inverse image(notice, the right upward arrow is local ring map)

□

◇ **Lemma 1.2.4.2** Let X be a locally ringed space, $f \in \Gamma(X, \mathcal{O}_X)$, the set

$$D(f) := \{x \in X \mid \text{image } f \notin m_x\}$$

is open. Moreover $f|_{D(f)}$ has an inverse.

◦ **Proof** Firstly, $x \in X \Leftrightarrow \rho_x(f) \notin m_x \Leftrightarrow \rho_x(f)$ is invertible in $\mathcal{O}_{X,x} \Leftrightarrow \text{Exists } g \in \Gamma(X, \mathcal{O}_X), \rho_x(f)\rho_x(g) = \rho_x(fg) = id_{\mathcal{O}_{X,x}} \Leftrightarrow \rho_x(fg - id_{\mathcal{O}_X(X)}) = 0 \Leftrightarrow \text{Exists } g, U, f|_U \circ g|_U = id_U$. And notice that $x \in U \subseteq D(f)$, so $D(f)$ is open.

Moreover, $f|_{D(f)}$ has an inverse: Suppose that f has inverse on U, V with $g, h \in \Gamma(X, \mathcal{O}_X)$:

$$g|_U \circ f|_U = f|_U \circ g|_U = id_U \quad h|_V \circ f|_V = f|_V \circ h|_V = id_V$$

Then on $U \cap V$, we have

$$g|_{U \cap V} = h|_{U \cap V}$$

With the pasting property of $\mathcal{O}_X$, there exists some $f^{-1} \in \Gamma(X, \mathcal{O}_X)$ and $f^{-1}|_{D(f)} \circ f|_{D(f)} = f|_{D(f)} \circ f^{-1}|_{D(f)} = id_{D(f)}$. $\square$

Moreover, in the chapter of invertible modules, we have the following conclusion in general:

◊ **Lemma 1.2.4.3** In lemma above, if $X = Spec(A)$ is an affine scheme, then, for $f \in A$,

$$D(f) = \{p \in Spec(A) | f \notin p\}$$

agrees with the definition previously.

◦ **Proof** $X = Spec(A)$, $x = p \in Spec(A)$, then: $\rho_x : A \rightarrow A_p, t \mapsto \frac{t}{1}$. We have

$$\rho_x(t) = \frac{t}{1} \notin m_x = pR_p \Leftrightarrow \frac{t}{1} \in (R - p)R_p$$

Suppose $\frac{t}{1} = \frac{a}{b}, a, b \in R - p$, then $tb = a$. But $t \in p \Rightarrow a \in p$ contradiction. So $t \notin p$. Conversely also hold, because $(R - p)R_p \cap pR_p = \emptyset$ with invertibility. So $\Leftrightarrow f \notin p$. $\square$

► **Theorem** Let

- X be a locally ringed space;
- Y be an affine scheme;

The map

$$\begin{aligned} Mor(X, Y) &\rightarrow Hom(\Gamma(Y, \mathcal{O}_Y), (\Gamma(X, \mathcal{O}_X))) \\ f &\mapsto f^\# \end{aligned}$$

is bijective.

◦ **Proof** Y is affine. Let $Y = Spec(R)$, then

→ Omitted

← Construction of the inverse map: Given some

$$\psi_Y : \Gamma(Y, \mathcal{O}_Y) = R \rightarrow \Gamma(X, \mathcal{O}_X)$$

Then, we will construct the corresponding map $\Psi : X \rightarrow Y$. Given $x \in X$, then define

$$\Psi(x) := \text{Ker}(\Gamma(Y, \mathcal{O}_Y) \xrightarrow{\psi_Y} \Gamma(X, \mathcal{O}_X) \longrightarrow \mathcal{O}_{X,x} \longrightarrow \mathcal{O}_{X,x}/m_x)$$

- Continuity: Take standard open $D(g) \subseteq Y = \text{Spec}(R)$. Then,

$$\Psi^{-1}(D(g)) = D(\Psi_Y(g))$$

which is open.

- Sheaf Structure Homomorphism: We should construct the Ψ -map

– Given by:

$$\Gamma(D(g), \mathcal{O}_Y) = \Gamma(Y, \mathcal{O}_Y)_g \rightarrow \Gamma(\Psi^{-1}(D(g)), \mathcal{O}_X)$$

And $\Psi_Y(g)$ is invertible on $D(\Psi_Y(g))$. Thus we could extend Ψ_Y to

$$\psi_{D(g)} : \Gamma(Y, \mathcal{O}_Y)_g \rightarrow \Gamma(\Psi^{-1}(D(g)), \mathcal{O}_X)$$

- The morphism is local for stalks. Suppose the morphism is defined as

$$(\Psi, \psi) : (X, \mathcal{O}_X) \rightarrow (Y, \mathcal{O}_Y)$$

Now,

$$\psi_x : \mathcal{O}_{Y, \Psi(x)} \rightarrow \mathcal{O}_{X,x}$$

it is local: 1.2.4.1

- Finally, there exists an inverse between $(\Psi, \psi) \mapsto \psi_Y$ and the construction $\psi_Y \mapsto (\Psi, \psi)$.

□

★ **Corollary 1.2.4.4** The category of affine schemes is equivalent to the opposite of the category of rings. The equivalence is given by the functor that associates to an affine scheme the global sections of its structure sheaf.

- **Proof** Omitted.

□

★ **Corollary 1.2.4.5** $\text{Spec}(\mathbb{Z})$ is the final object in the category of affine scheme/locally ringed space.

◦ **Proof** Suppose that $(X, \mathcal{O}_X)$ be a locally ringed space. Then, we have $\text{Hom}(X, \text{Spec}(\mathbb{Z})) \cong \text{Hom}(\Gamma(\text{Spec}(\mathbb{Z}), \text{Spec}(\mathbb{Z})), \Gamma(X, \mathcal{O}_X)) = \text{Hom}(\mathbb{Z}, \mathcal{O}_X(X))$. And the morphism $\mathbb{Z} \rightarrow \mathcal{O}_X(X)$ is well-defined.

□

1.2.5 Product, Fibre Product

◇ **Lemma 1.2.5.1** The category of affine schemes has:

- Finite products;
- Fibre products;

In other words, it has finite limits.

Moreover, the products and fibre products in the category of affine schemes are the same as in the category of locally ringed spaces. In formula

$$\begin{aligned} \text{Spec}(R) \times \text{Spec}(S) &= \text{Spec}(R \otimes_{\mathbb{Z}} S) \\ \text{Spec}(A) \times_{\text{Spec}(R)} \text{Spec}(B) &= \text{Spec}(A \otimes_R B) \end{aligned}$$

- **Proof**

-
-
-

□

1.2.6 Disjointed Union

◇ **Lemma 1.2.6.1** Let X be a locally ringed space. Assume that:

$$X = U \sqcup V$$

where U, V are affine opens. Then X is an affine scheme.

◦ **Proof** Define:

$$R = \mathcal{O}_X(U) \times \mathcal{O}_X(V)$$

Then, there exists a canonical morphism of locally ringed spaces

...

□

1.2.7 Quasi-Coherent Sheaf on Affines

We will prove the equivalence of following categories

$$QCoh(\mathcal{O}_{Spec(R)}) \Leftrightarrow Mod - R$$

Where $QCoh(\mathcal{O}_X)$ is a subcategory of $Mod(\mathcal{O}_X)$.

◇ **Lemma 1.2.7.1** Let:

- $(X, \mathcal{O}_X) = (Spec(R), \mathcal{O}_{Spec(R)})$ be an affine scheme;
- M be an R -module.

Then, there exists an isomorphism between sheaf $\hat{M}$ associated to the R -module M and the sheaf $\mathcal{F}_M$ associated to the R -module M . This isomorphism is functorial in module M .

Moreover they are characterized by the following mapping property

$$Hom_{\mathcal{O}_X}(\hat{M}, \mathcal{F}) = Hom_R(M, \Gamma(X, \mathcal{F}))$$

for any sheaf of $\mathcal{O}_X$ -modules $\mathcal{F}$. Here a map $\alpha : \hat{M} \rightarrow \mathcal{F}$ corresponds to its effect on global sections.

◦ **Proof** ...

□

◇ **Lemma 1.2.7.2** Let $(X, \mathcal{O}_X) = (Spec(R), \mathcal{O}_{Spec(R)})$ be an affine scheme. There exists canonical isomorphisms

- $\widetilde{M \otimes_R N} = \widetilde{M} \otimes_{\mathcal{O}_{Spec(R)}} \widetilde{N}$;
- $\widetilde{T^n M} \cong T^n(\widetilde{M})$, $\widetilde{Sym^n(M)} \cong Sym^n(\widetilde{M})$, $\widetilde{\wedge^n(M)} = \wedge^n(\widetilde{M})$;

- If M is finitely presented, then

$$\mathrm{Hom}_{\mathcal{O}_X}(\tilde{M}, \tilde{N}) \cong \widetilde{\mathrm{Hom}_R(\tilde{M}, N)}$$

◦ **Proof ...**

□

◇ **Lemma 1.2.7.3** ...

◦ **Proof ...**

□

◇ **Lemma 1.2.7.4** ...

◦ **Proof ...**

□

► **Theorem** Let $(X, \mathcal{O}_X) = (\mathrm{Spec}(R), \mathcal{O}_{\mathrm{Spec}(R)})$ be affine scheme. The functors

$$M \mapsto \tilde{M} \quad \mathcal{F} \mapsto \Gamma(X, \mathcal{F})$$

give a rise to the equivalence of functors

$$\mathrm{Mod}(\mathcal{O}_X) \xLeftrightarrow{\sim} \mathrm{Mod}\text{-}R$$

◦ **Proof ...**

□

1.2.8 Kernels, Cokernels

◇ **Lemma 1.2.8.1** Let $X = \mathrm{Spec}(R)$ be an affine scheme. Kernels and cokernels of maps of quasi-coherent $\mathcal{O}_X$ -modules are quasi-coherent.

◦ **Proof**

□

1.2.9 Direct Sum

◇ **Lemma 1.2.9.1** ...

◦ **Proof ...**

□

1.2.10 Exact Sequence

◇ **Lemma 1.2.10.1** Let $(X, \mathcal{O}_X) = (\mathrm{Spec}(R), \mathcal{O}_{\mathrm{Spec}(R)})$ be an affine scheme. Suppose that

$$0 \rightarrow \mathcal{F}_1 \rightarrow \mathcal{F}_2 \rightarrow \mathcal{F}_3 \rightarrow 0$$

is a short exact sequence of sheaves of $\mathcal{O}_X$ -modules. If two out of three are quasi-coherent so is the third.

◦ **Proof** ...

□

1.2.11 Open and Closed Subspaces of Affines

In general, the open/closed subspaces and immersions are discussed in locally ringed spaces.

Open Subspace

◇ **Lemma 1.2.11.1** Let Y be an affine scheme, $f \in \Gamma(Y, \mathcal{O}_Y)$, the open subspace $D(f)$ is an affine subspace.

◦ **Proof** As a matter of fact, $D(f)$ has a structure of ringed space by

$$\begin{aligned} \Gamma(D(f), \mathcal{O}_{D(f)}) &= \Gamma(Y, \mathcal{O}_Y)_f \\ \phi : U = \text{Spec}(R_f) &\rightarrow \text{Spec}(R) = Y \end{aligned}$$

We have know:

- $\text{Spec}(R_f) \rightarrow \text{Spec}(R)$ is a homeomorphism onto $D(f)$;
- $\phi^{-1}\mathcal{O}_Y \rightarrow \mathcal{O}_U$ is an isomorphism on stalk. Thus is an open immersion because isomorphism of sheaves.

□

Closed Subspace

♣ **Example 1.2.11.2** Let: R be a ring, $I \subseteq R$ be an ideal. Consider the morphism of affine schemes

$$i : Z = \text{Spec}(R/I) \rightarrow \text{Spec}(R) = X$$

This is a homeomorphism of Z onto the closed subset of X .

Moreover, if $I \subseteq p \subseteq R$ is a prime corresponding of a point $x = i(z)$, where $x \in X, z \in Z$, then on stalks we get the map

$$\mathcal{O}_{X,x} = R_p \rightarrow R_p/IR_p = \mathcal{O}_{Z,z}$$

Thus we see that i is a closed immersion of locally ringed spaces.

◇ **Lemma 1.2.11.3** Let $(X, \mathcal{O}_X) = (\text{Spec}(R), \mathcal{O}_{\text{Spec}(R)})$ be an affine scheme. Let $i : Z \rightarrow X$ be any closed immersion of locally ringed spaces. Then, there exists a unique ideal $I \subseteq R$ such that: $i : Z \rightarrow X$ can be identified with the closed immersion

$$\text{Spec}(R/I) \rightarrow \text{Spec}(R)$$

constructed above.

◦ **Proof** Such I is given by: The ideal sheaf $\mathcal{I} = \text{Ker}(\mathcal{O}_X \rightarrow i_*\mathcal{O}_Z)$ on $\text{Spec}(R)$. $I = \Gamma(\text{Spec}(R), \mathcal{I})$ is an ideal. (By kernel)

□

And such $\hat{\mathcal{I}}$ is naturally locally generated...

1.3 Topological Properties

More discussion should be listed in CA.

Chapter 2 Schemes

2.1 Definitions

Definition 2.1.0.1 A **scheme** is a locally ringed space $(X, \mathcal{O}_X)$, with

- Open Covering $X = \cup U_i$;
- $(U_i, \mathcal{O}_X|_{U_i})$ is affine scheme.

2.1.1 Morphisms of Schemes

Definition 2.1.1.1 The morphism of schemes is the morphism of locally ringed spaces.

Affinization Morphism

Definition 2.1.1.2 Define the functor:

$$aff : Sch \rightarrow Affsch \quad X \mapsto Spec(\mathcal{O}_X(X))$$

And induces a natural morphism $X \rightarrow Spec(\mathcal{O}_X(X))$

Especially,

$$(aff_X, aff_X^\#) : X \rightarrow Spec(\mathcal{O}_X(X))$$

- $\phi(x) = aff_X(x) := [\mathcal{O}_X(X) \rightarrow \mathcal{O}_{X,x}]^{-1}(m_x)$
- $aff_X(D(f)) : \mathcal{O}_X(X)_f \rightarrow \Gamma(\phi^{-1}(D(f)), \mathcal{O}_X)$ induced by the morphism $\mathcal{O}_X(X) \rightarrow f|D(f)$ is inversable.

Specified by the module homomorphism

$$\begin{aligned} Hom(X, Spec(\mathcal{O}_X(X))) &= Hom(\Gamma(Spec \mathcal{O}_X(X), \mathcal{O}_{Spec(\mathcal{O}_X(X))}), \Gamma(X, \mathcal{O}_X)) \\ &= Hom(\mathcal{O}_X(X), \mathcal{O}_X(X)) \end{aligned}$$

Take the identity.

◇ **Lemma 2.1.1.3** Firstly, aff is a covariant functor.

- **Proof** Consider morphism $\varphi \in Hom(X, Y)$. Then, we have $aff(\varphi) : Spec(\mathcal{O}_Y(Y)) \rightarrow Spec(\mathcal{O}_X(X))$. □

♠ **Proposition 2.1.1.4** The morphism $X \rightarrow Spec(A)$ factors through aff_X

$$X \rightarrow Spec(\mathcal{O}_X(X)) \rightarrow Spec(A)$$

- **Proof** Notice that: There exists $f^\#(Spec(A)) : A \rightarrow \mathcal{O}_X(X)$. We have correspondence $aff_X \Leftrightarrow id_{\mathcal{O}_X(X)}$. Then the factorization $Spec(\mathcal{O}_X(X)) \rightarrow Spec(A)$ is $Spec(f^\#(Spec(A)))$. Then consider the composition of $f^\#(Spec(A)) \circ aff_X^\# = f^\#$ □

♠ **Proposition 2.1.1.5** Scheme X is affine iff $af f_X$ is an isomorphism.

◦ **Proof** Omitted. □

Open Immersion

◇ **Lemma 2.1.1.6** Let X be a scheme, $j : U \rightarrow X$ be an open immersion of locally ringed spaces. Then, U is a scheme. In particular, any open subspace of X is a scheme.

◦ **Proof** Suppose $u \in U$. Then, select affine neighborhood $u \in V$, there must be some $u \in D(f) \subseteq U$. $D(f)$ is affine scheme, so U is a scheme via $U = \cup_{u \in U} D(f_u)$, with paasting of sheaves $\mathcal{O}_X|_{D(f_u)}$. □

★ **Corollary 2.1.1.7** Any scheme X has a basis for topology consisting of affine opens $D(f)$ where $f \in \Gamma(U, \mathcal{O}_X|_U) = (Spec(R), \mathcal{O}_{Spec(R)})$.

◦ **Proof** Omitted. □

Closed Immersion

◇ **Lemma 2.1.1.8** Let X be a scheme, $i : Z \rightarrow X$ be a closed immersion of locally ringed spaces:

- The locally ringed space Z is a scheme;
- The kernel $\mathcal{I}$ of the map $\mathcal{O}_X \rightarrow i_* \mathcal{O}_Z$ is a **quasi-coherent sheaf**;
- For any affine open $U = Spec(R)$ of X , the morphism

$$i^{-1}(U) \rightarrow U \quad Spec(R/I) \rightarrow Spec(R)$$

for some ideal $I \subseteq R$ and We have $\mathcal{I}|_U = \hat{I}$.

◦ **Proof** □

◇ **Lemma 2.1.1.9** Let $f : Y \rightarrow X$ be an immersion of schemes. Then f is closed immersion iff $f(Y) \subseteq X$ is a closed subset.(immersion: Here 2.1.1.10)

◦ **Proof** $\Rightarrow$ Omitted.

$\Leftarrow$ □

Immersion

Definition 2.1.1.10

- The morphism of schemes is called an **open immersion** if it is an open immersion of locally ringed spaces;
- An **open subscheme** of X is an open subspace of X is an open subspace of X or a scheme by taking subset;
- A morphism of schemes is called a **closed immersion** if it is a closed immersion of locally ringed spaces;

- A **closed subscheme** of X is a closed subspace of locally ringed space X or by lemma above;
- An **immersion** can be factored as $f = j \circ i$ where i is closed immersion, j is open immersion.

More contents of morphisms of schemes will be given in another chapter.

2.1.2 Zariski Topology of Schemes

◇ **Lemma 2.1.2.1** Let X be a scheme. A **irreducible closed subset** of X has a unique generic point. In other words, it is a sober space.

◦ **Proof** Select irreducible set Z

Existence of generic point: For every affine open $U = \text{Spec}(R)$ and we know that

$$Z \cap U = V(I)$$

for a unique radical ideal $I \subseteq R$. And $V(I)$ is irreducible $\Leftrightarrow I$ is prime. Thus $p = I$ is the generic point of $Z \cap U$ by $Z \cap U = V(I)$.

Now, suppose ξ' be another generic point, then there exists some neighborhood U contains ξ, ξ' by definition. Finally $\xi' = \xi$. □

◇ **Lemma 2.1.2.2** The underlying topological space of any scheme is locally quasi-compact.

◦ **Proof** Affine open is compact. Omitted. □

◇ **Lemma 2.1.2.3** Let X be a scheme, U, V be affine opens of X , $x \in U \cap V$.

There exists an affine open neighborhood W of x such that W is a standard open of both U and V .

◦ **Proof** Let $U = \text{Spec}(R)$ and $V = \text{Spec}(S)$. Then, suppose $p = q = x \in U \cap V$; $p \subseteq R, q \subseteq S$. □

◇ **Lemma 2.1.2.4** Let X be a scheme and $X = \bigcup U_i$ be an affine open covering. Then, if V is an affine open, there exists a standard open covering $V = \bigcup V_j$ where $V_j \subseteq U_i$ is a standard open.

◦ **Proof** ... □

◇ **Lemma 2.1.2.5** Let X be a scheme and $\mathcal{B}$ be the set of affine opens of X . Let $\mathcal{F}$ be a presheaf of sets on $\mathcal{B}$. The followings are equivalent:

- $\mathcal{F}$ is the restriction of sheaf on X to $\mathcal{B}$;
- $\mathcal{F}$ on $\mathcal{B}$ and
- $\mathcal{F}(\emptyset)$ is a singleton;
- Whenever $U = V \cup W$ with $U, V, W \in \mathcal{B}$ and $V, W \subseteq U$ standard open. the map

$$\mathcal{F}(U) \rightarrow \mathcal{F}(V) \times \mathcal{F}(W)$$

is injective with image the set of pair (s, t) such that $s|_{V \cap W} = t|_{V \cap W}$

◦ **Proof** ... □

Special Case: Scheme on Finite discrete Set

◇ **Lemma 2.1.2.6** Let X be a finite and discrete topological space and $(X, \mathcal{O}_X)$ be a scheme. Then it is affine.

◦ **Proof** Now consider $\coprod \{x_i\}$, then each x_i must be affine. Thus X is the finite disjoint union of affine schemes. Hence is affine by taking disjoint union. $\square$

♣ **Example 2.1.2.7** Existence of a scheme without closed point:

2.1.3 Points in Schemes

Definition 2.1.3.1 Given a scheme X we can define a functor

$$h_X : \text{Sch}^{opp} \rightarrow \text{Sets} \quad T \mapsto \text{Mor}(T, X)$$

This is called the **functor of points** of X .

Let

- X be a scheme;
- R be a local ring with maximal ideal m ;

Suppose that $f : \text{Spec}(R) \rightarrow X$ be a morphisms of schemes. Then, $x := f(m)$, we obtain a local homomorphism of local rings

$$f^\# : \mathcal{O}_{X,x} \rightarrow \mathcal{O}_{\text{Spec}(R),m} = R_m = R$$

Remark $R_m = R$. For local ring R , $R - m$ is the collection of invertible elements. Thus,

◇ **Lemma 2.1.3.2** Let X be a scheme, R be a local ring. Then, the construction above gives a bijective correspondence Morphisms $\text{Spec}(R) \rightarrow X \Leftrightarrow$ Pairs (x, φ) consisting of a point X and a local homomorphism of local rings $\mathcal{O}_{X,x} \rightarrow R$.

◦ **Proof** Firstly, we should consider the affine case: Let $X = \text{Spec}(A)$. Then, $\psi : A \rightarrow R$ has a factorization of local rings

$$A \rightarrow A_p \rightarrow R$$

where $p = \psi^{-1}(m)$. Thus $\text{Spec}(R) \rightarrow X$ is given by composition of $A \rightarrow A_p$ and $A_p \rightarrow R$, which is the local homomorphism $\mathcal{O}_{X,x} \rightarrow R$.

Now consider the general case. $\Leftarrow$ Consider $\mathcal{O}_X(U) \rightarrow \mathcal{O}_{X,x} \rightarrow R, \text{Spec}(R) \rightarrow U$ where U is affine. And paste the morphisms for sheaf.

Conversely, given $f : \text{Spec}(R) \rightarrow X$, we must prove: For any affine open of X such that $x = f(m) \in U, \text{Im}(f) \subseteq U$. In fact: $m \in f^{-1}(U)$, so $f^{-1}(U) = \text{Spec}(R)$. $\square$

As a special case of the lemma above we obtain for every point x of a scheme X a canonical morphism

$$\begin{array}{ccc} & \dots & \\ \text{Spec}(\mathcal{O}_{X,x}) & \longrightarrow & X \\ \downarrow & & \downarrow \\ \text{Spec}(\mathcal{O}_{S,s}) & \longrightarrow & S \end{array}$$

Generalization, Specialization

Definition 2.1.3.3

- Let $x, x' \in X$. If $x' \in \overline{\{x\}}$, we say that
 - x' is a specialization of x ;
 - x is a generalization of x'
- A subset $V \subseteq X$ is called **closed under specialization/generalization** if it contains all specializations (generalizations) of all its points.

◇ **Lemma 2.1.3.4** Let X be a scheme, $x, x' \in X$. Then:

$x' \in X$ is a generalization of $x \Leftrightarrow x'$ is in the image of canonical morphism $\text{Spec}(\mathcal{O}_{X,x}) \rightarrow X$.

◦ **Proof ...** □

♣ **Example 2.1.3.5** ...

Points-Spectra Field Correspondence

◇ **Lemma 2.1.3.6** Let X be a

◦ **Proof ...** □

Definition 2.1.3.7 Let $f : X \rightarrow Y$ be a continuous map, we say that

-
-

Valuative Criterion for Universal Closedness

Review: The **closed map** is the map which maps closed sets to closed sets.

Definition 2.1.3.8**Valuative Criterion of Separatedness****2.1.4 Representability**

Definition 2.1.4.1 Suppose that $F : \text{Sch}^{op} \rightarrow \text{Sets}$ be functor. Then, F is **representable** if there exists a scheme X such that $h_X \cong F$.

Now apply Yoneda's lemma we have

$$\text{Hom}(h_X, F) \cong F(X)$$

Thus $(X, s : h_X \rightarrow F)$ is unique up to unique isomorphism if it exists.

Moreover, we have a bijection

$$\text{Mor}_{\text{Fun}(\text{Sch}^{op}, \text{Sets})}(h_Y, F) \rightarrow F(Y) \quad s \mapsto s(id_Y)$$

Conversely, we have

$$F(Y) \rightarrow \text{Mor}_{\text{Fun}(\text{Sch}^{op}, \text{Sets})}(h_Y, F) \quad \xi \mapsto s_\xi$$

where $s_\xi : h_Y \rightarrow F$, $h_Y(T) = \text{Mor}(T, Y) \rightarrow F(T)$, $f \mapsto f^*\xi$

Remark In particular, in the case that F is representable, there exists a scheme X and an element $\xi \in F(X)$. We say

Definition 2.1.4.2 Pair (X, ξ) represents F , if $F = s_\xi$.

Definition 2.1.4.3 Such $\xi \in F(X)$ is often called the **universal family** for reasons that will become more clear when we talk about [algebraic stacks].

Then, every element $\xi' \in F(T)$ for any T is of the form $\xi' \in F(T)$ for any T is of the form $\xi' = f^*\xi$ for a unique morphism $f : T \rightarrow X$.

♣ **Example 2.1.4.4** Consider the rule which associates to every scheme T the set

$$F(T) = \Gamma(T, \mathcal{O}_T)$$

Then, we can turn this into a contravariant functor by using for a morphism $f : T \rightarrow T'$ the pullback $f^* : \Gamma(T', \mathcal{O}_{T'}) \rightarrow \Gamma(T, \mathcal{O}_T)$.

Thus, given a ring R with element $t \in R$ there exists a unique ring homomorphism $\mathbb{Z}[x] \rightarrow R$ which maps x to t , thus we have a bijection $R \cong \text{Hom}(\mathbb{Z}[x], R)$,

$$\text{Mor}(T, \text{Spec}(\mathbb{Z}[x])) = \text{Hom}(\mathbb{Z}[x], \Gamma(T, \mathcal{O}_T)) = \Gamma(T, \mathcal{O}_T)$$

This gives an isomorphism of functors

$$F \rightarrow h_{\text{Spec}(\mathbb{Z}[x])}$$

Criterion for Representability

Definition 2.1.4.5 Let F be a contravariant functor on category of schemes with values in sets:

- We say F satisfies the sheaf property for the Zariski topology if for every scheme T and every open covering $T = \cup_{i \in I} U_i$ and for any collection of elements $\xi_i \in F(U_i)$ such that

$$\xi_i|_{U_i \cap U_j} = \xi_j|_{U_i \cap U_j}$$

There exists a unique element $\xi \in F(T)$ such that $\xi_i = \xi|_{U_i} \in F(U_i)$;

- A **subfunctor** $H \subseteq F$ is a rule associate to every scheme T a subset

$$H(T) \subseteq F(T)$$

And for $T \rightarrow T'$ we have $F(f) : F(T) \rightarrow F(T')$; $H(f) = F(f)|_{H(T)} : H(T) \rightarrow H(T')$.

- Let $H \subseteq F$ be a subfunctor. We say $H \subseteq F$ is **representable by open immersions** if for all pairs (T, ξ) , where T is a scheme and $\xi \in F(T)$ there exists an open subscheme $U_\xi \subseteq T$ with following properties

A morphism $f : T' \rightarrow T$ factors through U_ξ iff $f^*\xi \in H(T')$.

- Let I be a set. For each $i \in I$ $H_i \subseteq F$ be a subfunctor. We say that the collection $(H_i)_{i \in I}$ covers F iff for every $\xi \in F(T)$ there exists an open covering $T = \cup U_i$ such that $\xi|_{U_i} \in H_i(U_i)$

◇ **Lemma 2.1.4.6** Let F be a covariant functor on the category of schemes with value in the category of sets. Suppose that:

- F satisfies the sheaf property for the Zariski topology;
- There exists a set I and a collection of subfunctors such that:
 - each F_i is representable;

- each $F_i \subseteq F$ is representable by open immersions;
- the collection $(F_i)_{i \in I}$ covers F

Then, F is representable.

◦ **Proof ...** □

Remark Suppose the functor F is defined on all locally ringed spaces, and if conditions of above are replaced by the following:

- F satisfies the sheaf property on the category of locally ringed spaces;
- There exists a set I and a collection of subfunctors $F_i \subseteq F$ such that:
 - Each F_i is representable by a scheme;
 - Each $F_i \subseteq F$ is representable by open immersions on the category of locally ringed spaces; and
 - The collection $(F_i)_{i \in I}$ cover F as a functor on the category of locally ringed spaces.

2.2 Classification of Schemes(by Rings)

2.2.1 Reduced Scheme

2.2.2 Integral, Irreducible and Reduced Scheme

Integral Scheme

Definition 2.2.2.1 Let X be a scheme. Then X is **integral** if for every nonempty affine open $\text{Spec}(R) = U$, R is an integral domain.

Irreducible Scheme

Definition 2.2.2.2 Scheme X is **irreducible** iff its underlying topological space $|X|$ is irreducible.

◇ **Lemma 2.2.2.3** The followings are equivalent:

- Scheme X is irreducible;
- There exists an affine open covering $X = \cup_i U_i$ such that I is not empty, U_i is irreducible for all $i \in I$. And $U_i \cap U_j \neq \emptyset$ for all $i, j \in I$;
- The scheme X is nonempty and every nonempty affine open $U \subseteq X$ is irreducible;

◦ **Proof ...** □

Integral $\Leftrightarrow$ Reduced + Irreducible

♠ **Proposition 2.2.2.4** A scheme X is integral iff it is reduced and irreducible.

◦ **Proof $\Rightarrow$**
 $\Leftarrow$ □

2.2.3 Types of Schemes defined by Properties of Rings

Local Properties

Definition 2.2.3.1

- Let P be a local property of rings. We say P is local if the following hold:
 -
 -
- Let X be a scheme. We say X is locally P if for any $x \in X$ there exists an affine open neighborhood U of x in X such that $\mathcal{O}_X(U)$ has property P .

◇ Lemma 2.2.3.2 ...

◦ Proof ...

□

◇ Lemma 2.2.3.3 ...

◦ Proof ...

□

◇ Lemma 2.2.3.4 The following properties of a ring R are local:

-
-
-
-

◦ Proof ...

□

2.2.4 Noetherian Schemes

2.2.5 Jacobson Schemes

Definition 2.2.5.1 ...

2.2.6 Normal Schemes

Definition 2.2.6.1 A scheme X is normal iff for all $x \in X$ the local ring $\mathcal{O}_{X,x}$ is a normal domain.

2.3 Fibre Products of Schemes

2.3.1 Schemes as Functors

Here: 2.1.3

2.3.2 Fibre Products

Existence of Fibre Products of Schemes

◇ **Lemma 2.3.2.1** The category of schemes has a

- Final Objects;
- Products;
- Fibre Products;

In other words, the category of schemes has finite limits.

◦ **Proof** Review: 1.2.4.5

Let $f : X \rightarrow S, g : Y \rightarrow S$ be morphisms of schemes. We have to show that the functor

$$F : Sch \rightarrow Sets \quad T \mapsto Mor(T, X) \times_{Mor(T, S)} Mor(T, Y)$$

is representable. We should apply the lemma: □

Another proof through affine case:

◇ **Lemma 2.3.2.2** (Affine Case) Suppose that $X = Spec(A), Y = Spec(B)$ with $S = Spec(C)$, with
Then, $X \otimes_S Y$ exists in Sch/S , and is given by $Spec(A \otimes_C B)$ together with the ring morphism

$$\begin{aligned} A &\rightarrow A \otimes B & a &\mapsto a \otimes 1 \\ B &\rightarrow A \otimes B & b &\mapsto 1 \otimes b \end{aligned}$$

◦ **Proof** ... □

★ **Corollary 2.3.2.3** For affine X, Y, S and morphisms $f : X \rightarrow S, g : Y \rightarrow S$ then $X \times_S Y$ is affine.

◦ **Proof** Omitted. □

Fibre Products

Definition 2.3.2.4 Given morphisms of schemes $f : X \rightarrow S$ and $g : Y \rightarrow S$, the **fibre product** is a scheme $X \times_S Y$ together with projection morphisms

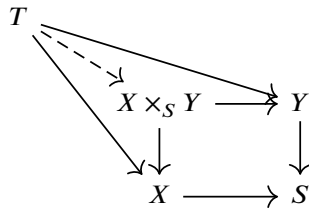
$$p : X \times_S Y \rightarrow X \quad q : X \times_S Y \rightarrow Y$$

With the following commutative diagram.

$$\begin{array}{ccc} X \times_S Y & \xrightarrow{q} & Y \\ \downarrow p & & \downarrow g \\ X & \xrightarrow{f} & S \end{array}$$

which is universal among all diagrams of this sort.

In other word, for $T \rightarrow X, T \rightarrow Y$, we have



Open Subschemes

♠ **Proposition 2.3.2.5** Let $f : X \rightarrow S$ and $g : Y \rightarrow S$ be morphisms of schemes with the same target, and $X \times_S Y, p, q$ be fibre product.

Suppose that $U \subseteq S, V \subseteq X, W \subseteq Y$ be open subschemes such that $f(V) \subseteq U, g(W) \subseteq U$. Then the canonical morphism

$$V \times_U W \rightarrow X \times_S Y$$

is an open immersion which identifies $V \times_U W$ with $p^{-1}(V) \cap q^{-1}(W)$.

◦ **Proof ...** □

♠ **Proposition 2.3.2.6** There exists a correspondence of points

$$X \times_S Y \ni z \Leftrightarrow (x, y, s, p)$$

where:

-
-

◦ **Proof**

$$X \times_S Y$$

$$\begin{array}{ccc}
 \text{Spec}() & \text{Spec} & Y \\
 \text{Spec}() & \text{Spec} & \\
 X & & S
 \end{array}$$

□

Points in Fibre Product

♠ **Proposition 2.3.2.7** ...

◦ **Proof ...** □

Immersiones

◇ Lemma 2.3.2.8 ...

◦ Proof ...

□

2.3.3 Base Change

Base Change

Definition 2.3.3.1 Let S be a scheme:

- We say X is a **scheme over S** to mean that X is equipped with a morphism of schemes

$$X \rightarrow S$$

The morphism $X \rightarrow S$ is sometimes called the **structure morphism**;

- If R is a ring we say X is a **scheme over S** if X is a scheme over $\operatorname{Spec}(R)$;
- We denote $\operatorname{Mor}_S(X, Y)$ the set of all morphisms from X to Y over S ,

$$\begin{array}{ccc} X & \xrightarrow{\quad} & Y \\ & \searrow \quad \swarrow & \\ & S & \end{array}$$

- (Base Change of Scheme)
- (Base Change of Morphisms)
-
-

Immersiones

◇ Lemma 2.3.3.2 ...

◦ Proof ...

□

Properties

Definition 2.3.3.3 ...

Scheme Theoretic Fibre

Definition 2.3.3.4 ...

◇ Lemma 2.3.3.5 ...

◦ Proof ...

□

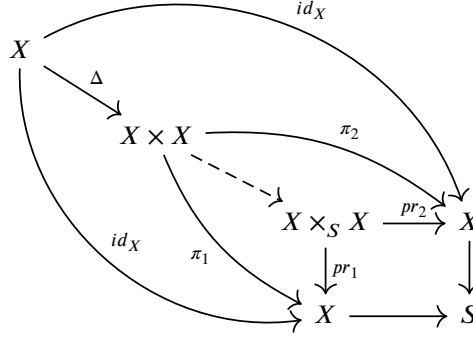
2.3.4 Separation Axioms

'Hausdorffness for Schemes'

Definition 2.3.4.1 A topological space X is Hausdorff iff $\Delta \subseteq X \times X$ is a closed subset.
 In analogue in AG is: Given a scheme X over S

$$\Delta_{X/S} : X \rightarrow X \times_S X$$

This is a unique morphism given by



And $\Delta_{X/S} : X \xrightarrow{\Delta} X \times X \xrightarrow{p} X \times_S X$ is the unique morphism such that

$$pr_1 \circ \Delta_{X/S} = pr_2 \circ \Delta_{X/S}$$

◇ **Lemma 2.3.4.2** The diagonal morphisms of a morphism between affines is closed.

◦ **Proof ...**

□

◇ **Lemma 2.3.4.3** Let X be a scheme over S . The diagonal morphism $\Delta_{X/S}$ is an immersion.

◦ **Proof ...**

□

Definition 2.3.4.4 Let $f : X \rightarrow S$ be a morphism of schemes:

- We say f is **separated** if the diagonal morphism $\Delta_{X/S}$ is a closed immersion;
- f is **quasi-separated** if the diagonal morphism $\Delta_{X/S}$ is quasi-compact morphism;
- We say scheme Y is **separated** if the morphism $Y \rightarrow \text{Spec}(Z)$ is separated;
- We say scheme Y is **quasi-separated** if the morphism $Y \rightarrow \text{Spec}(Z)$ is quasi-separated;

◇ **Lemma 2.3.4.5** ...

◦ **Proof ...**

□

2.4 Quasi-Coherent Modules

2.5 Schemes over Fields

2.5.1 Schemes over a Field which is not Algebraically Closed

title

Construction

2.6 Local Properties of Schemes

2.6.1 The Tangent Space

Formal Derivatives

2.7 Affine and Proper Morphisms

Affine Morphisms: 2.1.1.2

Chapter 3 Constructions

3.1 Glueing Scheme, Relative Glueing, Relative Spectrum

3.1.1 Glueing Scheme

Definition 3.1.1.1 (Glueing Data) Let I be a set. $(X_i, \mathcal{O}_i)$ be the locally ringed space. For each pair $i, j \in I$ and $U_{ij} \subseteq X_i$ be open subspace, and

$$\varphi_{ij} : U_{ij} \rightarrow U_{ji}$$

◇ **Lemma 3.1.1.2** ...

◦ **Proof** ...

□

3.1.2 Relative Spectrum via Glueing

Here:

- S is a scheme;
- $\mathcal{A}$ is a quasi-coherent $\mathcal{O}_S$ -algebra;

Relative Glueing

◇ **Lemma 3.1.2.1** Let S be a scheme, $\mathcal{B}$ be a basis for the topology of S . Suppose given the following data:

- For every $U \in \mathcal{B}$ a scheme

$$f_U : X_U \rightarrow U$$

over U of S ;

- For $U, V \in \mathcal{B}$ a morphism $\rho_V^U : X_V \rightarrow X_U$ over U ;

Assuming:

- Each ρ_V^U induces an isomorphism

$$X_V \rightarrow f_U^{-1}(V)$$

- Whatever $W, V, U \in \mathcal{B}$, with $W \subseteq V \subseteq U$ have

$$\rho_W^U = \rho_V^U \circ \rho_W^V$$

Then, there exists a morphism $f : X \rightarrow S$ of schemes and isomorphism

$$i_U : f^{-1}U \rightarrow X_U$$

over U such that for $V, U \in \mathcal{B}$, $V \subseteq U$ the composition

$$X_V \xrightarrow{i_V^{-1}} f^{-1}(V) \xrightarrow{\text{inclusion}} f^{-1}(U) \xrightarrow{i_U} X_U$$

is the morphism ρ_V^U . Moreover X is unique up to unique isomorphism over S .

◦ **Proof** Firstly, we set

$$F(T) := \{(g, \{h_U\}_{U \in \mathcal{B}}) \mid g : T \rightarrow S, h_U : g^{-1}(U) \rightarrow X_U, f_U \circ h_U = g|_{g^{-1}(U)}, h_U|_{g^{-1}(V)} = \rho_V^U \circ h_V, V \subseteq U\}$$

And we use the sheaf representability lemma:

- Sheaf Property:
 - – Representability
 - Representable by Open Immersion:
 - The collections (F_i) covers F :

□

◇ **Lemma 3.1.2.2** Let S be a scheme, $\mathcal{B}$ be a basis for the topology of S . Suppose given the following data:

- For every $U \in \mathcal{B}$ a scheme

$$f_U : X_U \rightarrow U$$

over U .

- For every $U \in \mathcal{B}$ a quasi-coherent sheaf $\mathcal{F}_U$ over X_U ;
- For every $U, V \in \mathcal{B}$, morphism $\rho_W^U = \rho_V^U \circ \rho_W^V$;
- For every $U, V \in \mathcal{B}$, morphism $\theta_V^U : (\rho_V^U)^* \mathcal{F}_U \rightarrow \mathcal{F}_V$

assume that:

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-
-
-

Then, there exists a morphism of schemes $f : X \rightarrow S$ together with a quasi-coherent sheaf on X and isomorphism $i_U : f^{-1}(U) \rightarrow X_U$; $\theta_U : i_U^* \mathcal{F}_U \rightarrow \mathcal{F}|_{f^{-1}(U)}$ and composition for $V \subseteq U$

...

◦ **Proof ..**

□

Affine Case

We will construct a scheme X over S : Especially, if $S = \text{Spec}(R)$ is affine, and $A = \Gamma(S, \mathcal{A})$, then define:

$$\underline{\text{Spec}}_S(\mathcal{A}) := \text{Spec}(A)$$

With sheaf morphism induced by $R \rightarrow A$.

$$\pi : \text{Spec}(A) = \underline{\text{Spec}}_S(\mathcal{A}) \rightarrow S = \text{Spec}(R)$$

Furthermore: Suppose $\mathcal{A}$ be a quasi-coherent $\mathcal{O}_S$ -algebra, then we can construct the morphism of schemes

$$\underline{Spec}_S(\mathcal{A}) \rightarrow S$$

by glueing $Spec(\Gamma(U, \mathcal{A}))$ over the affine opens of S .

◇ **Lemma 3.1.2.3** Suppose $U \subseteq U' \subseteq S$ be affine opens, and

$$A = \mathcal{A}(U), A' = \mathcal{A}(U')$$

The map of rings $A' \rightarrow A$ induces a morphism $Spec(A) \rightarrow Spec(A')$, and the diagram

$$\begin{array}{ccc} Spec(A) & \longrightarrow & Spec(A') \\ \downarrow & & \downarrow \\ U & \longrightarrow & U' \end{array}$$

is cartesian.

◦ **Proof ...**

□

◇ **Lemma 3.1.2.4** In situation above, suppose $U \subseteq U' \subseteq U'' \subseteq S$ are affine opens, $A = \mathcal{A}(U), A' = \mathcal{A}(U'), A'' = \mathcal{A}(U'')$, then the composition of morphisms $Spec(A) \rightarrow Spec(A'')$

◦ **Proof ...**

□

◇ **Lemma 3.1.2.5** There exists a morphism of schemes

$$\pi : \underline{Spec}_S(\mathcal{A}) \rightarrow S$$

with following properties:

-
-

◦ **Proof ...**

□

3.1.3 Relative Spectrum as Functor

Short review: For $f : T \rightarrow S$, there exists a pullback for quasi-coherent sheaf of $\mathcal{O}_S$ -algebras $\mathcal{A}$:

$$f^* \mathcal{A} \in \mathcal{O}_T - \text{Algs}$$

Consider pairs $(f : T \rightarrow S, \varphi)$ with corresponding morphism

$$\varphi : f^* \mathcal{A} \rightarrow \mathcal{O}_T$$

$$\varphi : f^{-1} \mathcal{A} \rightarrow \mathcal{O}_T$$

$$\varphi : \mathcal{A} \rightarrow f_* \mathcal{O}_T$$

Via $a : T' \rightarrow T$ we could get a second pair $(f' = f \circ a, \varphi' = a^* \varphi)$, which we call the **pullback**

◇ **Lemma 3.1.3.1** Let F be the functor associated to $(S, \mathcal{A})$ above, let $g : S' \rightarrow S$ be a morphism of schemes, $\mathcal{A}' = g^* \mathcal{A}$, let F' be the functor associated to $(S', \mathcal{A}')$ above, there exists a canonical isomorphism

$$F' \cong h_{S'} \times_{h_S} F$$

of functors.

◦ **Proof ...** □

◇ **Lemma 3.1.3.2** Let F be the functor associated to $(S, \mathcal{A})$, if S is affine, then F is representable by the affine scheme $\text{Spec}(\Gamma(S, \mathcal{A}))$.

◦ **Proof ...** □

◇ **Lemma 3.1.3.3** The functor

◦ **Proof ...** □

◇ **Lemma 3.1.3.4** ...

◦ **Proof ...** □

3.1.4 Relative Spectrum

Definition 3.1.4.1 Consider $\pi : \text{Spec}_S(\mathcal{A}) \rightarrow S$, the "universal family" is a family of $\mathcal{O}_S$ -algebras

$$\mathcal{A} \rightarrow$$

3.2 Affine n -Space

Definition 3.2.0.1 Let S be a scheme and $n \geq 0$. Define:

$$A_S^n = (\mathcal{O}_S[T_1, \dots, T_n])$$

called **affine n -space over S** .

If $S = \text{Spec}(R)$ is affine, then this affine n -space over R and we denote it A_R^n .

Base Change: Especially, let $S \rightarrow S'$ of schemes we have

$$g^* \mathcal{O}_S[T_1, \dots, T_n] = \mathcal{O}_{S'}[T_1, \dots, T_n]$$

and hence $A_{S'}^n = S' \times_S A_S^n$ is the base change. Therefore an alternative definition of affine n -space is the formula

$$A_S^n = S \times_{\text{Spec} \mathbb{Z}} A_{\mathbb{Z}}^n$$

Also, a morphism from an S -scheme $f : X \rightarrow S$ to A_S^n is given by a homomorphism of $\mathcal{O}_S$ -algebras

$$\mathcal{O}_S[T_1, \dots, T_n] \rightarrow f_* \mathcal{O}_X$$

with the bijection $\text{Hom}_{\text{Sch}/S}(X, \text{Spec}_S(\mathcal{A})) \cong \text{Hom}_{\mathcal{O}_S\text{-alg}}(\mathcal{A}, f_* \mathcal{O}_X)$ We could define the value elements $h_i \in \Gamma(X, \mathcal{O}_X)$.

3.3 Vector Bundles

Definition 3.3.0.1 Let S be a scheme and $\mathcal{E}$ be a quasi-coherent $\mathcal{O}_S$ -module. The **vector bundle** associated to $\mathcal{E}$ is

$$V(\mathcal{E}) = \underline{\text{Spec}}_S(\text{Sym}(\mathcal{E}))$$

Then, we have the grading

$$\pi_* \mathcal{O}_{V(\mathcal{E})} = \bigoplus_{n \geq 0} \text{Sym}^n(\mathcal{E})$$

Definition 3.3.0.2 Let S be a scheme. A **vector bundle** $\pi : V \rightarrow S$ over S is an affine morphism of schemes such that

3.3.1 Vector Bundle and Locally Free Bundles

3.3.2 Torsors and Non-Abelian Cohomology

3.3.3 Vector Bundles and GL_n -Torsors

,

3.3.4 Flattening Stratification for Modules

3.3.5 Divisors

3.4 Cones

3.5 Proj

3.5.1 Proj of a graded Ring(Algebra)

3.5.2 Standard Open Covering

◇ **Lemma 3.5.2.1** Let S be a graded ring, and $f \in S$ with $\deg(f) > 0$:

- If $g \in S$ homogeneous of positive degree with $D_+(g) \subseteq D_+(f)$, then:
 - f is invertible in S_g ; $\frac{f^{\deg(g)}}{g^{\deg(f)}}$ is invertible in $S_{(g)}$;
 - $g^e = af$ for some $e \geq 1$ and $a \in S$ homogeneous;
 - there is a canonical S -algebra map $S_f \rightarrow S_g$;
 - there is a canonical S_0 -algebra map $S_{(f)} \rightarrow S_{(g)}$ compatible with the map $S_f \rightarrow S_g$;
 - The map $S_{(f)} \rightarrow S_{(g)}$ induces an isomorphism

$$(S_{(f)})_{g^{\deg(f)}/f^{\deg(g)}} \cong S_{(g)}$$

- These maps induce a commutative diagram of topological spaces

...

- There are compatible canonical S_f and $S_{(f)}$ -module maps

$$M_f \rightarrow M_g \quad M_{(f)} \rightarrow M_{(g)}$$

induces an isomorphism

$$(M_{(f)})_{g^{\deg(f)}/f^{\deg(g)}} = M_{(g)}$$

- Any open covering of $D_+(f)$ can be refined to be a finite open covering of the form

$$D_+(f) = \cup_{i=1}^n D_+(g_i)$$

- Let $g_1, \dots, g_n \in S$ be homogeneous of positive degree. Then,

$$D_+(f) \subseteq \cup D_+(g_i) \Leftrightarrow g_1^{deg(f)} / f^{deg(g_1)}, \dots, g_n^{deg(f)} / f^{deg(g_n)} \text{ generate the unit ideal in } S(f)$$

◦ **Proof ...**

□

Definition 3.5.2.2 Let S be a graded ring. Suppose that $D_+(f) \subseteq Proj(S)$ is a standard open.

A **standard open covering** of $D_+(f)$ is a covering $D_+(f) = \cup_{i=1}^n D_+(g_i)$ where $g_1, \dots, g_n \in S$ are homogeneous of positive degree.

3.5.3 Structure Sheaf

3.5.4 Quas-Coherent Sheaves on $Proj$

3.5.5 Invertible Sheaves on $Proj$

3.5.6 Functoriality of $Proj$

3.5.7 Morphisms into $Proj$

3.6 Projective Space

3.7 Relative $Proj$

3.8 Projective Bundles

3.9 Grassmannians

Chapter 4 Classification of Schemes

4.1 Types of Schemes by Rings

4.1.1 Local

Definition 4.1.1.1

- Let P be a property of rings. We say that P is **local** if the followings hold

- For any ring R , any $f \in R$ we have

$$P(R) \Rightarrow P(R_f)$$

- For any ring R and $f_i \in R$, $R = (f_1, \dots, f_n)$, then

$$\forall i, P(R_{f_i}) \Rightarrow P(R)$$

- Let P be a property of rings, X be a scheme. We say X is locally P , if for any $x \in X$ there exists an affine open neighbourhood $x \in U$ such that $\mathcal{O}_X(U)$ has the property P .

◇ **Lemma 4.1.1.2** Let X be a scheme, and P is a local property of rings. The following are equivalent:

-
-
-
-

◦ **Proof ...**

□

◇ **Lemma 4.1.1.3** Let X be a scheme. Then X is reduced iff X is locally reduced.

◦ **Proof ...**

□

◇ **Lemma 4.1.1.4** The following properties of a ring R are local:

- (Cohen-Macaulay)
- (Regular)
- (Absolutely Noetherian)
-

◦ **Proof ...**

□

4.2 Reduced Scheme

Definition 4.2.0.1 Let X be a scheme. We say X is **reduced** if every local ring $\mathcal{O}_{X,x}$ is reduced.

Reduced on Sections and on Stalks

◇ **Lemma 4.2.0.2** A scheme X is reduced iff $\mathcal{O}_X(U)$ is a reduced ring for all open $U \subseteq X$ open.

◦ **Proof** $\Rightarrow$ Let $f \in \mathcal{O}_X(U)$ such that $f^n = 0$. Then, $f^n|_u = 0, \forall u \in U \Rightarrow f|_u = 0, \forall u \in U$. Thus $f = 0$.
 $\Leftarrow$ Conversely, if for all $\mathcal{O}_X(U)$, it is reduced. Then let $\tilde{f} \in \mathcal{O}_{X,x}$ be nonzero with nonzero representative element $f \in \mathcal{O}(U)$. Thus $f^n \neq 0 \Rightarrow \tilde{f}^n \neq 0$ □

Affine Scheme

◇ **Lemma 4.2.0.3** An affine scheme $\text{Spec}(R)$ is reduced iff R is reduced.

◦ **Proof** $\Rightarrow$ Take $\mathcal{O}_{\text{Spec}(R)}(\text{Spec}(R))$;
 $\Leftarrow$ If R is reduced, then $\mathcal{O}_{\text{Spec}(R),p} = R_p$ is reduced. □

4.2.1 Scheme Structure on Closed Subset

◇ **Lemma 4.2.1.1** Let X be a scheme, $T \subseteq X$ be a closed subset. There exists a unique closed subscheme $Z \subseteq X$ with the following properties:

- The underlying topological space of subscheme $|Z| = T$;
- Z is reduced.

◦ **Proof**

- Firstly, consider the subscheme of X by

$$\mathcal{I}(U) := \{f \in \mathcal{O}_X(U) | \forall t \in T \cap U, f(t) = 0\}$$

where $f(t) := \kappa(t)$. $\mathcal{I}$ is a sheaf on X :

- Identity: As a substructure of X ;
- Pasting: Given $s_i \in \mathcal{I}(U_i)$. Then we will prove the pasting $s \in \mathcal{I}(U)$: Omitted.

Moreover, take the open affine $U = \text{Spec}(R)$ we have

$$T \cap U = V(I)$$

for a unique radical ideal $I \subseteq R$. Now given $p \in V(I)$ corresponding to $t \in T \cap U$ and an element . Hence:

$$\mathcal{I}(U) = \cap_{p \in V(I)} p = I$$

Moreover, take standard open $D(g) \subseteq \text{Spec}(R) = U$ we have

$$\mathcal{I}(D(g)) = I_g$$

Thus $\mathcal{I}$ is quasi-coherent sheaf of ideals.

- Now take the closed subspace Z associated to I . For every affine open $U = \text{Spec}(R)$ of X we see that

$$Z \cap U = \text{Spec}(R/I)$$

where I is a radical ideal, so Z is reduced. And the underlying topological space $|Z| = \text{supp}(\mathcal{O}_X/I) = T$.

- Uniqueness: Now let $Z' \subseteq X$ be a second closed subscheme with the properties. But notice that on every affines, we have $V(I) = V(I')$ because of uniqueness of I or I' we have $Z = Z'$.

□

Definition 4.2.1.2 Let X be a scheme, $Z \subseteq X$ be a closed subset.

A **scheme structure** on Z is given by a closed subscheme Z' of X whose underlying set is equal to Z .

Especially, the scheme structure on X could be denoted as X_{red} .

Remark For locally closed subset T , then T is a closed subset of the open subscheme $X - \partial T$ of X , hence $\partial T = \bar{T} - T$ is the boundary of T in the topological space X .

(T is closed in $\bar{T}$, by definition)

◇ **Lemma 4.2.1.3** Let X be a scheme. $Z \subseteq X$ be a closed subscheme. Y be reduced scheme. The morphism $f : Y \rightarrow X$ factors through $Z \Leftrightarrow f(Y) \subseteq Z$.

In particular, any morphism $Y \rightarrow X$ factors as

$$Y \rightarrow X_{\text{red}} \rightarrow X$$

◦ **Proof** $\Rightarrow$ Omitted;

$\Leftarrow$ Assume $f(Y) \subseteq Z$, then

□

◇ **Lemma 4.2.1.4** The followings are equivalent

-
-

◦ **Proof** ...

□

4.3 Integral, Irreducible and Reduced Schemes

4.3.1 Integral Scheme

4.3.2 Irreducible Scheme

4.4 Noetherian Schemes

Definition 4.4.0.1

- We say X is **locally Noetherian** if for every $x \in X$ has an affine open neighborhood $\text{Spec}(R) = U$ such that ring R is Noetherian;
- We say X is Noetherian if X is locally Noetherian and quasi-compact;

◇ **Lemma 4.4.0.2** (Equivalent Conditions for Noetherian Scheme) Let X be a scheme, the following are equivalent:

- The scheme X is locally Noetherian;
- For every affine open $U \subseteq X$ the ring $\mathcal{O}_X(U)$ is Noetherian;
- There exists an affine open covering $X = \cup U_i$ such that each $\mathcal{O}_X(U_i)$ is Noetherian;
- There exists an affine covering $X = \cup X_j$ such that each X_j is locally Noetherian.

Moreover, if X is locally Noetherian then every open subscheme is locally Noetherian.

◦ **Proof ...** □

Immersion

◇ **Lemma 4.4.0.3** Any immersion $Z \rightarrow X$ with X locally Noetherian is quasi-compact.

◦ **Proof ...** □

◇ **Lemma 4.4.0.4**

◦ **Proof ...** □

Underlying Space

◇ **Lemma 4.4.0.5** A (locally) Noetherian scheme has a (locally) Noetherian underlying topological space.

◦ **Proof ...** □

Morphism with Source Noetherian

◇ **Lemma 4.4.0.6** ...

◦ **Proof ...** □

Closed Point

Specialization

◇ **Lemma 4.4.0.7** Let X be a locally Noetherian scheme. Let $x' \rightsquigarrow x$ be a specialization of points of X . Then:

- There exists a discrete valuation ring R and a morphism

$$f : \operatorname{Spec}(R) \rightarrow X$$

such that the generic point η of $\operatorname{Spec}(R)$ maps to x' and the special point maps to x ; and

- provided $x \neq x'$ given a finitely generated field extension $K/\kappa(x')$, we may arrange it so that the extension $\kappa(\eta)/\kappa(x')$ induced by f is isomorphic to the given one.

◦ **Proof ...** □

◇ **Lemma 4.4.0.8** Let S be a Noetherian scheme. Let $T \subseteq S$ be an infinite subset. Then, there exists an infinite subset $T' \subseteq T$ such that there are no nontrivial specialization among the points T' .

◦ **Proof ...** □

◇ **Lemma 4.4.0.9** ...

◦ **Proof ...** □

Countable Dense Subset

◇ **Lemma 4.4.0.10** Let S be a Noetherian scheme, let $T \subseteq S$ be an infinite dense subset, then there exists a countable subset $E \subseteq T$ which is dense in S .

◦ **Proof ...** □

4.5 Jacobson Schemes

Definition 4.5.0.1

- Let X be a topological space, let X_0 be the set of closed points of X . We say: X is **Jacobson** if every closed subset $Z \subseteq X$ is the closure of

$$Z \cap X_0$$

- A scheme S is said to be **Jacobson** if its underlying topological space is Jacobson.

◇ **Lemma 4.5.0.2** An affine scheme $\text{Spec}(R)$ is Jacobson iff the ring R is Jacobson.

◦ **Proof ...** □

Equivalent Conditions

◇ **Lemma 4.5.0.3** Let X be a scheme. The following are equivalent

- The scheme X is Jacobson;
- The scheme is 'locally Jacobson'; (In the sense of 'local property' of Schemes by Rings)
- For every affine open $U \subseteq X$ the ring $\mathcal{O}_X(U)$ is Jacobson.
- There exists an affine open covering $X = \cup U_i$ such that each $\mathcal{O}_X(U_i)$ is Jacobson;
- There exists an open covering $X = \cup X_j$ such that each open subscheme X_j is Jacobson.

◦ **Proof ...**

□

Noetherian Jacobson Schemes

◇ **Lemma 4.5.0.4**

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-
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◦ **Proof ...**

□

4.6 Normal Schemes

Definition 4.6.0.1 A scheme X is normal iff for all $x \in X$, the local ring $\mathcal{O}_{X,x}$ is a normal domain.

4.7 Cohen-Macaulay Schemes

4.8 Regular Schemes

4.9 Dimension of Schemes

Definition 4.9.0.1 Let X be a scheme:

- The dimension of X is just the dimension of X as a topological space;
- For $x \in X$ we denote

$$\dim_x(X) := \min\{\dim(U) \mid x \in U \text{ open}\}$$

◇ **Lemma 4.9.0.2** The followings are equal:

- The dimension of X ;
- The supremum of the dimensions of local rings of X ;
- The supremum of $\dim_x(X)$ for $x \in X$;

◦ **Proof**

- -
 -
-

□

◇ **Lemma 4.9.0.3** Let X be a scheme, $Y \subseteq X$ be an irreducible closed subset. Let $\xi \in Y$ be the generic point. Then

$$\text{codim}(Y, X) = \dim(\mathcal{O}_{X, \xi})$$

where the codimension is as defined in topology.

◦ **Proof ...**

□

◇ **Lemma 4.9.0.4** Let X be a scheme with $x \in X$. Then x is a generic point of an irreducible component of X iff $\dim(\mathcal{O}_{X, x}) = 0$

◦ **Proof ...**

□

4.9.1 Catenary Schemes

Definition 4.9.1.1

- A ring A is called **catenary** if for any pair of prime ideals $p \subseteq q$ there exists a maximal chain of primes

$$p = p_0 \subseteq \cdots \subseteq p_e = q$$

and all of these have the same length.

- Let S be a scheme. We say S is **catenary** if the underlying topological space of S is catenary.

◇ **Lemma 4.9.1.2** Let S be a scheme, the following are equivalent:

- S is catenary
- There exists an open covering of S all of whose members are catenary;
- for every affine open $\text{Spec}(R) = U \subseteq S$ the ring R is catenary;
- There exists an affine open covering $S = \cup U_i$ such that each U_i is the spectrum of a catenary ring;
- For any $x \in X$, the local ring $\mathcal{O}_{X, x}$ is catenary.

◦ **Proof ...**

□

Locally Noetherian Rings

◇ **Lemma 4.9.1.3** Let S be a locally Noetherian scheme. The following are equivalent:

- S is catenary;
- locally in the Zariski topology there exists a dimension function on S ;

◦ **Proof ...**

□

4.9.2 Serre's Condition

Definition 4.9.2.1 Let X be a locally Noetherian scheme, $k \geq 0$:

- We say X is regular in codimension k , or we say X has the property (R_k) if for every $x \in X$ we have

$$\dim(\mathcal{O}_{X,x}) \leq k \Rightarrow \mathcal{O}_{X,x} \text{ is regular}$$

- We say X has property (S_k) if for every $x \in X$ we have

$$\text{depth}(\mathcal{O}_{X,x}) \geq \min(k, \dim(\mathcal{O}_{X,x}))$$

4.9.3 The Singular Locus

4.10 Japanese and Nagata Schemes

4.11 Quasi-Affine Scheme

Definition 4.11.0.1 ...

4.12 Ample Invertible Sheaves

Review Given an invertible sheaf $\mathcal{L}$ on a locally ringed space X , and given a global section of $\mathcal{L}$, the set

$$X_s := \{x \in X \mid s \notin m_x \mathcal{L}_x\}$$

is open (Consider as open subscheme). Take the following remark

$$X_{ss'} := X_s \cap X_{s'}$$

where ss' denote the section $s \otimes s' \in \Gamma(X, \mathcal{L} \otimes \mathcal{L}')$.

Definition 4.12.0.1 Let X be a scheme, $\mathcal{L}$ be an invertible $\mathcal{O}_X$ -module. We say $\mathcal{L}$ is **ample** if

- X is quasi-compact;
- for every $x \in X$ there exists an $n \geq 1$ and $s \in \Gamma(X, \mathcal{L}^{\otimes n})$ such that $x \in X_s$ and X_s is affine.

◇ **Lemma 4.12.0.2** Let X be a scheme, $\mathcal{L}$ be an invertible $\mathcal{O}_X$ -module, $n \geq 1$. Then, $\mathcal{L}$ is ample iff $\mathcal{L}^{\otimes n}$ is ample.

◦ **Proof** ...

□

◇ **Lemma 4.12.0.3** Let X be a scheme, $\mathcal{L}$ be an invertible $\mathcal{O}_X$ -module. Let $s \in \Gamma(X, \mathcal{L})$, for any affine $U \cap X$ the intersection

$$U \cap X_s$$

is affine.

◦ **Proof** ...

□

Tensor Product

◇ **Lemma 4.12.0.4** Let X be a scheme, let $\mathcal{L}$ and $\mathcal{M}$ be invertible $\mathcal{O}_X$ -modules. If

-
-

◦ **Proof ...**

□

Topology

4.12.1 Quasi-Coherent Sheaves and Ample Invertible Sheaves

Situation Let X be a scheme. Let $\mathcal{L}$ be an ample invertible sheaf on X . Then, set

$$S := \Gamma_*(X, \mathcal{L}) = \bigoplus_{n \geq 0} S_n, S_n := \Gamma(X, \mathcal{L}^{\otimes n})$$

Set: $Y := \text{Proj}(S)$, then $f : X \rightarrow Y$ be the canonical morphism

4.13 Appendix: How to find Suitable Affine Opens

Chapter 5 Morphism of Schemes

5.1 Classification of Properties

This section is just an elementary introduction.

5.1.1 Properties of Morphisms

Notation Let $f : X \rightarrow Y, g : Y' \rightarrow Y$. Now let

$$X_{Y'} := X \times_Y Y' \quad f_{Y'} : X_{Y'} \rightarrow Y'$$

$$\begin{array}{ccc} X_{Y'} & \xrightarrow{f_{Y'}} & Y' \\ \downarrow g' & & \downarrow g \\ X & \xrightarrow{f} & Y \end{array}$$

Definition 5.1.1.1 Let P be a property of morphisms of schemes

- (COMP) It is stable under composition. That is:

For every (pair) $f : X \rightarrow Y, g : Y \rightarrow Z$ satisfies $P \Rightarrow g \circ f$ satisfies P ;

- (CANC) It is cancellable. That is :

For all $f : X \rightarrow Y, g : Y \rightarrow Z; g \circ f$ satisfies $P \Rightarrow f$ satisfies P

- (BC) It is stable under base change,

For all $f : X \rightarrow Y, g : Y' \rightarrow Y$ satisfies $P \Rightarrow f_{Y'} : X_{Y'} \rightarrow Y'$.

- (LOCT) It is (Zariski-)local on the target, if for all morphism $f : X \rightarrow Y$, all open covers $Y = \cup_i V_i$, the morphism f satisfies P iff $f|_{V_i} : f^{-1}(V_i) \rightarrow V_i$ satisfies P for all $i \in I$.
- (LOCS) It is (Zariski-)local on the source if for all morphism $f : X \rightarrow Y$ and all open covers $X = \cup U_i$, the morphism f satisfies P iff $f|_{U_i}$ satisfies P for all $i \in I$.
- (Universally) A morphism $f : X \rightarrow Y$ is **universally** P if for all morphism $g : Y' \rightarrow Y$, morphism $f_{Y'}$ satisfies P .

Remark: Universally In algebra, we define

Definition 5.1.1.2 Ring homomorphism $\phi : A \rightarrow B$ is universally P : For any ring homomorphism $A \rightarrow C$, the induced homomorphism $\phi_C : C \rightarrow B \otimes_A C$ satisfies P

$$\begin{array}{ccc} B \otimes_A C & \longrightarrow & C \\ \downarrow & & \downarrow \\ B & \longrightarrow & A \end{array}$$

♣ **Example 5.1.1.3** Property 'flatness; is universal.

◇ **Lemma 5.1.1.4** Let $f : X \rightarrow Y$ be a morphism of schemes over S . Then: The following square is Cartesian:

$$\begin{array}{ccc}
 X & \xrightarrow{\Gamma_f} & X \times_S Y \\
 \downarrow f & & \downarrow f_Y \\
 Y & \xrightarrow{\Delta_{Y/S}} & Y \times_S Y
 \end{array}$$

◦ **Proof ...**

□

Definition 5.1.1.5 If P is a property of morphisms of schemes. We say that a morphism satisfies Δ_P if $\Delta_{X/Y}$ satisfies P .

♠ **Proposition 5.1.1.6** ...

◦ **Proof ...**

□

◇ **Lemma 5.1.1.7** ...

◦ **Proof ...**

□

► **Theorem** (Cancellation Theorem)

5.1.2 Topological Properties

Definition 5.1.2.1 A morphism $(f, f^\#) : (X, \mathcal{O}_X) \rightarrow (Y, \mathcal{O}_Y)$ of schemes:

- is inj./surj./bi. if f is inj./surj./bi.;
- is open if f is;
- is quasi-compact if f is quasi-compact;
- is quasi-separated if $\Delta_{X/Y} : X \rightarrow X \times_Y X$ is quasi-compact;
- has finite fibers if $f^{-1}(y)$ is finite for all $y \in Y$;

♠ **Proposition 5.1.2.2** The properties satisfies:

-
-
-
-
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-
-
-

◦ **Proof**

-
-
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-
-

□

5.1.3 Scheme-Theoretic Properties

Definition 5.1.3.1

- (Locally Closed) Immersion
- Locally of finite type if: For all affine open
- of finite type:
- quasi-finite if:
- affine: If
- Finite:
- Separated:
- Proper: If

5.2 Valuative Criteria

Definition 5.2.0.1 ...

5.3 Quasi-Compact Morphism

Definition 5.3.0.1 A morphism $f : X \rightarrow S$ is **quasi-compact** if its underlying map of topological space is quasi-compact.
i.e. for compact $K \subseteq S$ we have $f^{-1}(K)$ is compact.

♠ **Proposition 5.3.0.2** Let $f : X \rightarrow S$ be a morphism of schemes. The followings are equivalent:

- $f : X \rightarrow S$ is quasi-compact;
 - the inverse image of every affine open is quasi-compact;
-

- there exists some affine open covering $S = \cup U_i$ such that $f^{-1}(U_i)$ is quasi-compact for all i ;

◦ **Proof ...**

□

♠ Proposition 5.3.0.3

Closed Immersion

◇ **Lemma 5.3.0.4** A closed immersion is quasi-compact.

◦ **Proof ...**

□

Specialization

♠ **Proposition 5.3.0.5** Let $f : X \rightarrow S$ be a quasi-compact morphism of schemes. The followings are equivalent

- $f(X) \subseteq S$ is closed;
- $f(X) \subseteq S$ is stable under specialization;

◦ **Proof ...**

□

♠ **Proposition 5.3.0.6** Suppose $f : X \rightarrow S$ be a quasi-compact morphism of schemes. Then, f is closed iff specialization lift along f .

◦ **Proof ...**

□

5.4 Universal Closedness

Definition 5.4.0.1 A morphism of schemes $f : X \rightarrow S$ is said to be **universally closed** if every base change

$$f' : X_{S'} \rightarrow S'$$

is closed.

5.4.1 Valuative Criterion

5.5 Immersion

5.5.1 Open Immersion

5.5.2 Closed Immersion

5.5.3 Closed Immersions and Quasi-Coherent Sheaves

5.5.4 Support

5.6 Scheme Theoretic Image

◇ **Lemma 5.6.0.1** Let $f : X \rightarrow Y$ be a morphism of schemes. There exists a closed subscheme $Z \subseteq Y$ such that f

factors through Z , and that for any other closed subscheme $Z' \subseteq Y$ such that f factors through Z' we have $Z \subseteq Z'$.

◦ **Proof** Define $\mathcal{I} := \text{Ker}(\mathcal{O}_Y \rightarrow f_*\mathcal{O}_X)$, then by earlier discussion we have known, that $\mathcal{I}$ is quasi-coherent then we just take Z to be the closed subscheme determined by $\mathcal{I}$.

In general, to show that there exists a largest ideal $\mathcal{I}'$ contained in $\mathcal{I}$. □

Definition 5.6.0.2 ...

5.7 Scheme Theoretic Closure and Density

5.8 Quasi-Compact Morphisms

Review: A map $f : X \rightarrow Y$ is called **quasi-compact** iff the preimage $f^{-1}(V)$ of every compact V is compact.

Definition 5.8.0.1 A morphism of schemes is called **quasi-compact** if the underlying map of topological spaces is quasi-compact.

◇ **Lemma 5.8.0.2**

◦ **Proof** ... □

Quasi-Compact $\Rightarrow f(X)$ is Closed $\Leftrightarrow f(X)$ is stable under Spelization

◇ **Lemma 5.8.0.3** ...

◦ **Proof** ... □

Quasi-Compact $\Rightarrow f$ is closed $\Leftrightarrow$ Specialization lift along f

◇ **Lemma 5.8.0.4** ...

◦ **Proof** ... □

5.9 Monomorphisms

Definition 5.9.0.1 A morphism of schemes is called a **monomorphism** if it is a monomorphism in the category of schemes.

◇ **Lemma 5.9.0.2** ...

◦ **Proof** ... □

5.10 Surjective Morphisms

Definition 5.10.0.1 A morphism of schemes is said to be **surjective** if it is surjective on underlying topological spaces.

◇ **Lemma 5.10.0.2** The composition of surjective morphisms is surjective.

◦ **Proof** Omitted. □

5.10.1 Properties

Fibre Product

◇ **Lemma 5.10.1.1** Let X, Y be schemes over a base schemes S . Then, given points $x \in X, y \in Y$, there is a point of $X \times_S Y$ mapping to x and y under the projection iff x, y lie above the same point of S .

◦ **Proof** ... □

The Base Change

◇ **Lemma 5.10.1.2** ...

◦ **Proof** ... □

5.11 Radicial and Universally Injective Morphisms

Definition 5.11.0.1 ...

5.12 Dominant Morphisms

Definition 5.12.0.1 A morphism $f : X \rightarrow S$ of schemes is called **dominant** if the image of f is dense subset of S .

♣ **Example 5.12.0.2** Suppose k be an infinite field, $(\lambda_i)_{i \in \mathbb{N}}$ be the elements in k . Then, the morphism

$$\sqcup_i \text{Spec}(k) \rightarrow \text{Spec}(k[x])$$

induced by $x \mapsto \prod_i (\lambda_i)$

◇ **Lemma 5.12.0.3** Let $f : X \rightarrow S$ be a morphism of schemes, if every generic point of every irreducible component of S is in the image of f , then f is dominant.

◦ **Proof** ... □

◇ **Lemma 5.12.0.4** Let $f : X \rightarrow S$ be quasi-compact morphism of schemes.

◦ **Proof** ... □

5.12.1 Irreducible Component

5.12.2 Base Change

5.13 Affine Morphisms

Definition 5.13.0.1 A morphism of scheme $f : X \rightarrow S$ is called **affine** if the inverse image of every affine open of S is an affine open of X .

◇ **Lemma 5.13.0.2** Morphism $f : X \rightarrow S$ is affine $\Leftrightarrow f$ is separated and quasi-compact.

◦ **Proof ...**

□

5.13.1 Relative Spectrum

◇ **Lemma 5.13.1.1** ...

◦ **Proof ...**

□

◇ **Lemma 5.13.1.2** ...

◦ **Proof ...**

□

5.14 Families of Ample Invertible Modules

Definition 5.14.0.1 Let X be a scheme, $\{\mathcal{L}_i\}$ be a family of invertible $\mathcal{O}_X$ -modules. We say $\{\mathcal{L}_i\}$ is an **ample family of invertible modules** on X if:

- X is quasi-compact;
- for every $x \in X$, there exists an $i \in I$ and $n \geq 1$, and $s \in \Gamma(X, \mathcal{L}_i^{\otimes n})$ such that
 - $x \in X_s$;
 - X_s is affine;

5.15 Morphisms of Finite Type

5.16 Dimension

5.16.1 Morphisms and Dimension of Fibres

Definition 5.16.1.1 For $x \in X$ define

$$\dim_x(X) := \min\{\dim(U) \mid x \in U \text{ is open}\}$$

◇ **Lemma 5.16.1.2** Let $f : X \rightarrow S$ be a morphism of schemes, $x \in X$ and $s = f(x)$. Assume that f is locally of finite type, then

$$\dim_x(X_S) = \dim(\mathcal{O}_{X_{s,x}}) + \text{trdeg}_{\kappa(s)}(\kappa(x))$$

◦ **Proof ...**

□

5.16.2 Morphisms of Given Relative Dimension

Definition 5.16.2.1 ...

5.17 Ample

5.17.1 Relatively Ample Sheaves

5.17.2 Very Ample Sheaves

5.17.3 Ample and Very Ample Sheaves Relative to Finite Type Morphisms

5.18 Proper Morphisms

5.19 Valuative Criteria

5.20 Projective Morphisms

5.21 Integral and Finite Morphisms

5.22 Universal Homeomorphism

5.22.1 Definitions

5.22.2 Universal Homeomorphisms of Affine Schemes

5.23 Normalization

5.23.1 Absolute Weak Normalization and Seminormalization

5.23.2 Relative Normalization

5.23.3 Normalization

5.23.4 Weak Normalization

5.23.5 Weak Relative Noether Normalization

5.24 Finite Locally Free Morphisms

5.25 Rational Maps

5.25.1 Definitions

5.25.2 Birational Morphisms

5.25.3 Zariski's Main Theorem

5.26 Universally Bounded Fibres

5.27 Sections of Quasi-Coherent Sheaves

◇ **Lemma 5.27.0.1** Let A be a ring. Let $I \subseteq A$ be a finitely generated ideal. M be an A -module. Then, there is a canonical map

$$\operatorname{colim}_n \operatorname{Hom}_A(I^n, M) \rightarrow \Gamma(\operatorname{Spec}(A) - V(I), \hat{M})$$

The maps is always injective.

If for all $x \in M$, we have $Ix = 0 \Rightarrow x = 0$ then this map is an isomorphism.

In general, set $M_n := \{x \in M \mid I^n x = 0\}$, there exists an isomorphism

$$\operatorname{colim}_n \operatorname{Hom}_A(I^n, M/M_n) \rightarrow \Gamma(\operatorname{Spec}(A) - V(I), \hat{M})$$

◦ **Proof ...**

□

♣ **Example 5.27.0.2** The first displayed map is not an isomorphism.

Let k be a ring,

◇ **Lemma 5.27.0.3** Let X be a quasi-compact scheme, $\mathcal{I} \subseteq \mathcal{O}_X$ be a quasi-coherent sheaf of ideals of finite type. Let $Z \subseteq X$ be the closed subscheme defined by $\mathcal{I}$ with $U = X - Z$. Let $\mathcal{F}$ be a quasi-coherent $\mathcal{O}_X$ -module. The canonical map

$$\operatorname{colim}_n \operatorname{Hom}_{\mathcal{O}_X}(\mathcal{I}^n, \mathcal{F}) \rightarrow \Gamma(U, \mathcal{F})$$

is injective. Assume further that X is quasi-separated, let $\mathcal{F}_n \subseteq \mathcal{F}$ be subsheaf of sections annihilated by $\mathcal{I}^n$, the canonical map

$$\operatorname{colim}_n \operatorname{Hom}_{\mathcal{O}_X}(\mathcal{I}^n, \mathcal{F}/\mathcal{F}_n) \rightarrow \Gamma(U, \mathcal{F})$$

◦ **Proof ...**

□

5.28 Thickenings

5.28.1 Definitions

5.28.2 Morphisms

5.28.3 Picard Groups of Thickenings

5.28.4 Universal First Order Thickenings

5.29 Unramified Morphisms

5.29.1 Definitions

5.29.2 Formally Unramified Morphisms

5.30 Etale Morphisms

5.30.1 title

5.30.2 Formally Etale Morphisms

5.31 Smooth Morphisms

5.31.1 title

5.31.2 Formally Smooth Morphisms

5.31.3 Smoothness over a Noetherian Base

5.32 The

Chapter 6 Classification of Schemes

Chapter 7 Divisors

7.1 Associated Points

Definition 7.1.0.1 Let X be a scheme. Let $\mathcal{F}$ be a quasi-coherent sheaf on X :

- We say $x \in X$ is **associated to $\mathcal{F}$** if the maximal ideal m_x is associated to the $\mathcal{O}_{X,x}$ -module $\mathcal{F}_x$, i.e. There exists $a \in \mathcal{F}_x$

$$m_x = \text{Ann}(a)$$

- Take the following notation:

$$\text{Ass}(\mathcal{F}) \quad \text{or} \quad \text{Ass}_X(\mathcal{F})$$

- And: The associated points of X are associated points of $\mathcal{O}_X$.

Remark These definitions are most useful when X is locally Noetherian and $\mathcal{F}$ of finite type. For example it may happen that a generic point of an irreducible component of X is not associated to X , see Example 2.7. In the non-Noetherian case it may be more convenient to use weakly associated points, see Section 5. Let us link the scheme theoretic notion with the algebraic notion on affine opens; note that this correspondence works perfectly only for locally Noetherian schemes.

◇ **Lemma 7.1.0.2** Let X be a scheme. Let $\mathcal{F}$ be a quasi-coherent sheaf on X . Let $\text{Spec}(A) = U \subseteq X$ be an affine open, and set $M = \Gamma(U, \mathcal{F})$, $x \in U$, $p \subseteq A$ is the corresponding prime:

- If p is associated to M , then x is associated to $\mathcal{F}$;
- If p is finitely generated, then the converse holds as well.

In particular, if X is Noetherian, then the equivalence

$$p \in \text{Ass}(M) \Leftrightarrow x \in \text{Ass}(\mathcal{F})$$

◦ **Proof ...**

□

7.1.1 Weakly Associated Points

7.1.2 Morphisms and Weakly Associated Points

7.1.3 Support

◇ **Lemma 7.1.3.1** Let X be a scheme, $\mathcal{F}$ be a quasi-coherent $\mathcal{O}_X$ -module, then

$$\text{Ass}(\mathcal{F}) \subseteq \text{Supp}(\mathcal{F})$$

◦ **Proof ...**

□

Short Exact Sequence

Locally Noetherian Scheme

7.1.4 Morphisms and Associated Points

7.1.5 Embedded Points

7.2 Assassin

7.2.1 Relative Assassin

7.2.2 Relative Weak Assassin

7.3 Fitting Ideals

7.4 Effective Cartier Divisors

7.4.1 Definitions

7.4.2 Effective Cartier Divisors and

7.5 Norm

7.6 Weil Divisors

Chapter 8 Varities

8.1 Definitions

Definition 8.1.0.1 Let k be a field. A **variety** is a scheme X over k such that X is integral and the structure morphism

$$X \rightarrow \operatorname{Spec}(k)$$

is separated and of finite type.

8.1.1 Varieties and Rational Maps

8.2 Base Field Changes

Chapter 9 Topology of Schemes

Chapter 10 Limits of Schemes

Chapter 11 Chow Homology and Chern Classes

Chapter 12 Intersection Theory

Remark

12.1 Cycles

12.1.1 Definitions

Chapter 13 title